

Expt 7

Spectral Estimation

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

Objectives:

1. To generate a discrete-time signal consisting of sinusoidal components and noise.
2. To estimate the Power Spectral Density (PSD) of the signal using the Periodogram method.
3. To identify the frequency components present in the signal.
4. To compare the signal power and noise power for assessing the quality of generate signal.

Procedure:

1. Open MATLAB and create a new script file.
2. Define the sampling frequency and number of samples.
3. Generate a discrete-time signal consisting of two sinusoidal components.
4. Add white Gaussian noise to the signal.
5. Plot the generated signal in the time domain.
6. Use the periodogram() function to estimate the PSD.
7. Plot the PSD versus frequency.
8. Identify the dominant frequency components from the PSD peaks.
9. Analyze the spectral characteristics of the signal.
10. Verify that the detected frequencies match the original signal frequencies.

Program (Matlab code):

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Signal Parameters
```

```
fs = 1000;
```

```
t = (0:999)/fs;
```

```
%% Original Signal
```

```
s = sin(2*pi*100*t) + 0.5*sin(2*pi*200*t);
```

```
%% Noise
```

```
n = 0.3*randn(size(t));
```

```
%% Noisy Signal
```

```
x = s + n;
```

```
%% Figure 1 - Signal with Noise
```

```
figure;
```

```
plot(t, x);
```

```
grid on;
```

```
title('Signal with Noise');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
%% Figure 2 - PSD using Periodogram
```

```
figure;
```

```
periodogram(x, [], [], fs);
```

```
title('PSD using Periodogram');
```

```
%% Figure 3 - Magnitude Spectrum
```

```
figure;
```

```
X = abs(fft(x));
```

```
f = (0:length(X)-1)*fs/length(X);
```

```
plot(f(1:500), X(1:500));
```

```
grid on;
```

```
title('Magnitude Spectrum');
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
```

%% Figure 4 - Power Comparison

figure;

```
bar([mean(s.^2), var(n)]);
```

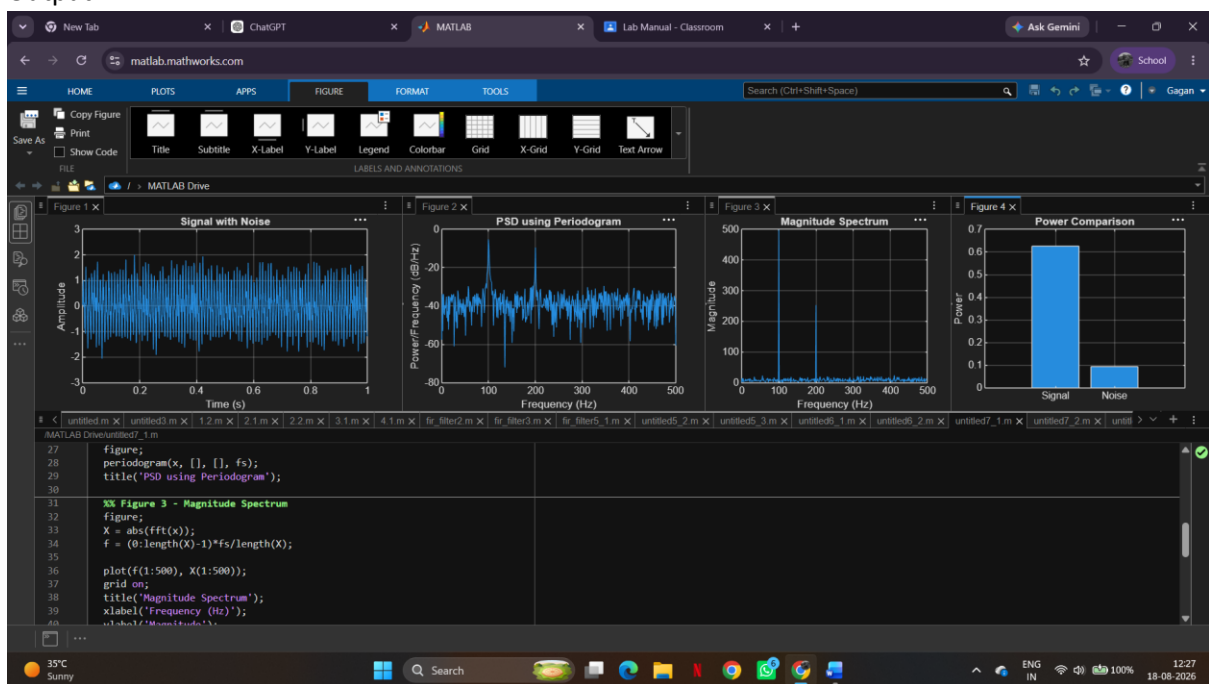
```
set(gca, 'XTickLabel', {'Signal', 'Noise'});
```

```
title('Power Comparison');
```

```
ylabel('Power');
```

```
grid on;
```

Output:



Result:

1. A discrete-time signal consisting of sinusoidal components with added noise was successfully generated in MATLAB.
2. The Power Spectral Density (PSD) of the signal was estimated using the Periodogram method.
3. The dominant frequency components present in the signal were accurately identified from the PSD plot.
4. The spectral characteristics of the signal were analyzed, showing the distribution of signal power across frequencies and the effect of noise on the spectrum.